#Milestone 1- Project Scope

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5W + 1H Analysis for Project Scope

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1. What:

Project Title:
"Securing Cloud-Based Web Servers using TLS Encryption, Traffic Tunneling, and Zero Trust Architecture"

Project Overview:
This project aims to design and deploy a secure cloud-based web server using **TLS encryption**, **Cloudflare Tunnel**, and **Zero Trust architecture**. The server will be hosted on a cloud platform, such as **AWS**, **Google Cloud**, or **DigitalOcean**, and will enforce strong encryption and authentication protocols. The traffic to the server will be tunneled securely through **Cloudflare Tunnel** to protect the origin server from direct exposure to the internet. Additionally, **Single Sign-On (SSO)** will be implemented via **Cloudflare Zero Trust**, ensuring that only authorized users can access sensitive areas of the application.

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2. Who:

Team Members:

- **Gautam Kumar**: **Lead Cloud Engineer**, responsible for setting up the cloud infrastructure and configuring the web server.
- **Devarshi Patel**: **Security Architect**, responsible for configuring TLS encryption, proxy traffic through Cloudflare Tunnel, and ensuring end-to-end security.
- **Saima Khan**: **Access Control Specialist**, tasked with implementing Cloudflare Zero Trust for access control, SSO integration, and securing sensitive routes.

Company: SecureSphere Technologies
Industry: SaaS, E-commerce, and Financial Services

Business Scenario:
SecureSphere Technologies is an innovative SaaS provider that specializes in creating cloud-based applications for the financial and e-commerce sectors. The company has seen rapid growth, particularly as it expands its clientele to include companies that process large volumes of sensitive financial and customer data. As part of this expansion, SecureSphere needs to guarantee that all customer transactions and internal processes are encrypted, secure, and fully compliant with **GDPR** and **PCI DSS** regulations.

The company has recently faced growing concerns regarding data breaches and unauthorized access to its systems, given the sensitive nature of the data handled (e.g., payment details, financial transactions, personal information). The company has decided to adopt a **Zero Trust** approach to its infrastructure, where no entity—internal or external—is trusted by default, and all access is verified and monitored.

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3. When:

Project Timeline:
The project is expected to be completed within a 10-week timeline:

- **Week 1-2**: Initial setup of the cloud-based web server using **AWS**, **Google Cloud**, or **DigitalOcean**. Basic configurations for server functionality and performance.
- **Week 3-4**: Implement **TLS encryption** to ensure all traffic is encrypted, using certificates from **Let's Encrypt** or **AWS Certificate Manager**.
- **Week 5-6**: Configure **Cloudflare Tunnel** to proxy traffic securely, hiding the origin server's IP from external access and adding DDoS protection.
- **Week 7-8**: Integrate **Single Sign-On (SSO)** using **Cloudflare Zero Trust**, allowing secure, role-based access control.
- **Week 9**: Conduct vulnerability testing and security checks to identify potential weaknesses.
- **Week 10**: Finalize the deployment, perform final tests, document the project, and provide recommendations for future security improvements.

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4. Why:

Business Need:
As SecureSphere Technologies continues to expand its services to high-profile clients in sectors such as financial services and e-commerce, the need for robust security has become a top priority. Clients expect their sensitive data to be protected at all times, with

stringent access controls in place. Failure to secure customer data could lead to severe financial penalties, loss of client trust, and damage to the company's reputation.

Technical Rationale:
To address these concerns, the project will implement **end-to-end encryption** using **TLS** for all communication, ensuring that data exchanged between clients and the server is secure. The project will also deploy **Cloudflare Tunnel** to mitigate the risks of DDoS attacks and unauthorized IP access by routing all traffic through Cloudflare's secure proxy infrastructure. Finally, by integrating **Cloudflare Zero Trust** and **SSO**, SecureSphere Technologies will enforce strict access controls, allowing only authenticated and authorized users to access sensitive areas of the system.

This approach will significantly improve SecureSphere's security posture, reducing the likelihood of data breaches and ensuring compliance with international data security regulations.

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5. Where:

Cloud Infrastructure:
The web server will be deployed on **AWS**, **Google Cloud**, or **DigitalOcean**, depending on which platform best aligns with the company's infrastructure and scalability needs. Traffic will be routed through **Cloudflare Tunnel**, which provides additional layers of security by masking the origin server's IP address and mitigating potential attacks. All services and communications will be encrypted with **TLS** to prevent unauthorized access or data interception.

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6. How:

Implementation Approach:

- **Cloud Setup**: Deploy the web server using AWS EC2, Google Compute Engine, or DigitalOcean Droplets. The server will be configured to serve HTTP requests and return HTTP headers in the response.
- **TLS Encryption**: Implement **SSL/TLS certificates** using Let's Encrypt or a certificate manager from AWS or Google Cloud, ensuring that all communication is encrypted.
- **Cloudflare Tunnel**: Set up Cloudflare Tunnel to proxy traffic from the client to the origin server, ensuring the server's IP is not exposed and traffic is securely routed.

- **Single Sign-On (SSO) & Zero Trust**: Configure **SSO** and **Cloudflare Zero Trust** to restrict access to certain areas of the application. Only authenticated users with the appropriate permissions will be able to access sensitive routes, enhancing security.
- **Monitoring & Testing**: Use tools such as **Cloudflare Analytics**, **AWS CloudWatch**, or **Google Cloud Monitoring** to track traffic patterns and identify any potential security risks. Security audits will be conducted to ensure the system is resilient against attacks.

Conclusion:

By implementing this security architecture, SecureSphere Technologies will not only protect its web applications from potential attacks but also ensure compliance with the latest security standards and regulations. This comprehensive solution—combining **TLS encryption**, **Cloudflare Tunnel**, and **Zero Trust architecture**—will position the company as a leader in secure SaaS solutions for high-risk industries, reinforcing client trust and securing long-term growth.